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Acoustics of Infant Pain Cries: Fundamental Frequency as a Measure of Arousal

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Abstract

Purpose: The goal of the current study is to examine fundamental frequency (F0) from a sample of healthy newborn infant pain cries.

Methods: A total of 58 healthy infants were included in the study. Data were examined from the first cry following a pain stimulus, in addition to a comparison of the first 30 seconds and the last 30 seconds of the entire crying episode.

Results: In the current study, the mean F0 was found to decrease over time (regardless of infant sex, term status, or positioning). The data also showed a significantly higher F0 for preterm female infants (compared to preterm males) in the first cry immediately following the pain stimulus.

Conclusions: These results reflect high levels of laryngeal tension following the pain stimulus, followed by an overall decrease in tension /arousal over time. The results support examination of the first cry following a pain stimulus, along with examination of the entire cry episode. Implications for the study of arousal and sudden infant death syndrome are discussed.

There is a relatively long history of using characteristics of infant cries to learn about an infant's health status. Some previous work has included perceptual analysis of cries, including studies of parental perception versus nonparental perception (Green, Jones, & Gustafson, 1987), correlation between perception and acoustics (e.g., Gustafson & Green, 1989; Protopapas & Eimas, 1997), and differences in perception related to health issues or risk for health issues (e.g., Lester, Boukydis, Garcia-Coll, Hole, & Peucker, 1992; Zeskind & Lester, 1978). Although perception of infant cries can be seen as important, a majority of work with infant cries has involved acoustic analysis of the cry signal.

A main focus of cry studies has been to use the acoustic signal to infer neurological or physiological status of the infant. Specifically, acoustic analysis of infant cry has have been used to differentiate healthy preterm infants from healthy full-term infants (Cacace, Robb, Saxman, Risemberg, & Koltai, 1995; Goberman & Robb, 1999; Johnston, Stevens, Craig, & Grunau, 1993; Lester, 1987; Robb, 2003; Thoden, Jarvenpaa, & Michelsson, 1985). In addition, acoustic analysis has been used to differentiate healthy infants from infants with respiratory problems (Hirschberg, 1990), laryngeal problems (Goberman & Robb, 2005), and neurological problems (Lind, Vuorenkoski, Rosberg, Partanen, & Wasz-Höckert, 1970). Further,

researchers have used infant cries to differentiate healthy infants from those with in utero exposure to cigarette smoke (Nugent, Lester, Greene, Wiczorek-Deering, & O'Mahony, 1996), marijuana (Lester & Dreher, 1989), cocaine (e.g., Corwin et al., 1992; Lester et al., 1991), and alcohol (Nugent et al., 1996; Zeskind, Platzman, Coles, & Schuetze, 1996). More recent work has examined healthy infants to attempt to infer position-related acoustic changes (Lin & Green, 2007), and position-related risk for sudden infant death syndrome (SIDS; Goberman, Johnson, Cannizzaro, & Robb 2008). Other studies have examined acoustic characteristics of infants who eventually died of SIDS (Colton & Steinschneider, 1981; Corwin et al., 1995; Stark & Nathanson, 1975). Overall, the acoustic cry signal has proven to be a useful indicator of physiological and/or neurological status of an infant.

Methodological Issues in Infant Cry Analysis

Types of Cries

Because cries can represent either volitional or reflexive responses to a stimulus, the acoustic and perceptual characteristics of cries can vary. Although there are acoustic differences between infant vocalization types (Fuller & Horii, 1986, 1988), pain cries have been used most often, given that they represent the most reflexive infant vocalization behavior. Elicitation techniques for pain cries can vary. Pain cries have been collected following standard painful medical procedures (Goberman & Robb, 1999, 2005; Goberman et al., 2008, Grunau & Craig, 1987; Johnston et al., 1993), or based on other painful stimuli such as rubber band snap (Zeskind et al., 1996), or heel flick (Corwin et al., 1992). Including pain cries, other vocalization techniques have been the basis for infant vocalization analysis, including cooing (Fuller & Horii, 1986, 1988), fussiness (Fuller, 1991), and hunger (Fairbanks, 1942; Gilbert & Robb, 1996).

Acoustic Measures

The earliest known analysis of crying used musical notation to describe infant cries (Gardiner, 1838), however more recently a number of different acoustic measures have been used to examine infant vocalizations. Analysis has focused on examining spectrographic features of cry (e.g., Lester, 1987; Robb & Cacace, 1995; Truby & Lind, 1965), and durational measures of cry episodes (Robb & Goberman, 1997; Robb, Goberman, & Cacace, 1997). Finally, long-term average spectra (LTAS) measures (Fuller & Horii, 1988; Goberman & Robb, 1999; Goberman et al., 2008), and cepstral analysis (Hariharan, Chee, & Yaacob, 2012) have been studied.

Across nearly all decades of infant cry research, fundamental frequency (F0) has been the most commonly examined measure (e.g., Fairbanks, 1942; Fuller & Horii, 1986; Gilbert & Robb, 1996; Truby & Lind, 1965; Wermke & Robb, 2010; Zeskind & Lester, 1978). Physiologically, F0 relates to the rate of vibration of the vocal folds, reflecting both tension and mass of the vocal folds. Infant cry F0 analyses have resulted in mixed findings. For example, Murry, Amundson, and Hollein (1977) found no F0 differences in pain cries versus other types of cries, whereas Fuller and Horii (1986) found pain cries to exhibit higher F0 compared to cooing or hunger cries, reflecting the increased arousal associated with infant pain. In addition, the analysis techniques used to measure F0 have varied. Some studies have sampled a limited number of individual cries for each infant, from only 2–3 cries (e.g., Corwin et al., 1992; Fuller, 1991), to 8–15 cries (Michelsson, Eklund, Leppänen, & Lyytinen, 2002), whereas other studies have examined full crying episodes (Goberman et al., 2008; Robb & Goberman, 1997; Robb et al., 1997; Wilder & Baken, 1977).

Summary

The goal of the current study was to examine F0 from a relatively large sample of healthy newborn infant pain cries. Researchers examined differences in F0 across different groups of healthy infants and examined F0 data from different periods of time within the crying

episode. Based on previous literature, a focus was placed on comparing F0 of preterm infants with full-term infants, along with a comparison of cries immediately following the pain stimulus to later cries. We expected that F0 would reflect the expectation of a maximum level of arousal immediately following the pain stimulus, followed by decreased levels of arousal at the end of the crying episode.

Methods

Participants

The participants in the current study were recruited from infants under the care of the Department of Obstetrics, Wood County Hospital, in Bowling Green, Ohio. The data collection procedures were approved by appropriate Human Subjects boards. A total of 58 infants were included in the study (34 males, 24 females), in addition to 11 preterm infants (<37 weeks gestational age; 7 males, 4 females) and 47 full-term infants (27 males, 20 females). Finally, 17 infants were placed in the prone position during recording (12 males, 5 females), whereas 40 infants were placed in the supine position (22 males, 18 females).

Overall, this sample of infants was healthy, with 91.4% of all Apgar scores being 8 or 9 at 1 minute, and 98.3% of all Apgar scores being 9 at 5 minutes of age. Previous research indicates that infants born in good condition achieve Apgar scores of 8 to 10, with an average of 8.4 (Finster & Wood, 2005). No infants had any medical diagnoses except being a newborn. Mean biological data were within normal limits of healthy typical newborn infants (weight=3394g, length=19.7in, head circumference=13.6in).

Procedures

Researchers collected data from infants during and immediately following a state-mandated metabolic screening procedure that involved lancing of the heel for a blood sample collection. For this group of infants, the blood sample was collected between 24 hours of age and the infant's discharge from the hospital (mean recording time=33.97 hours of age; SD=10.17 hours). The blood draw procedure was sufficiently painful to elicit crying behavior in the 58 infants studied. All infants were quiet and awake before the procedure, and all cries were recorded until the conclusion of the crying episode (defined as 10 seconds of non-cry behavior).

Cries were recorded with a constant mouth-microphone distance of 15 cm, using a Shure SM58 cardioid microphone coupled with a Marantz PMD430 audiocassette recorder. The microphone was positioned at mid-torso level and aimed at the infant's head for infants in both supine and prone positions. Recordings took place in a separate quiet room, situated adjacent to the nursery. Prior to analysis, all cries were digitized using CSL 4400 (2001), at 16kHz sampling rate with 16 bit quantization.

Researchers extracted the initial 30 seconds and final 30 seconds of the cry sample manually using the acoustic analysis software PRAAT (Boersma & Weenink, 2013). They completed three separate F0 measures using a cross-correlation method for pitch analysis. Researchers took measurements of mean F0 from the first cry, first 30 seconds, and final 30 seconds. They inspected pitch range, narrow-band spectrogram, and F0 contour to ensure the PRAAT-based pitch analysis was tracking F0 well. One infant was excluded from the analysis because the PRAAT-based pitch analysis did not validly track F0.

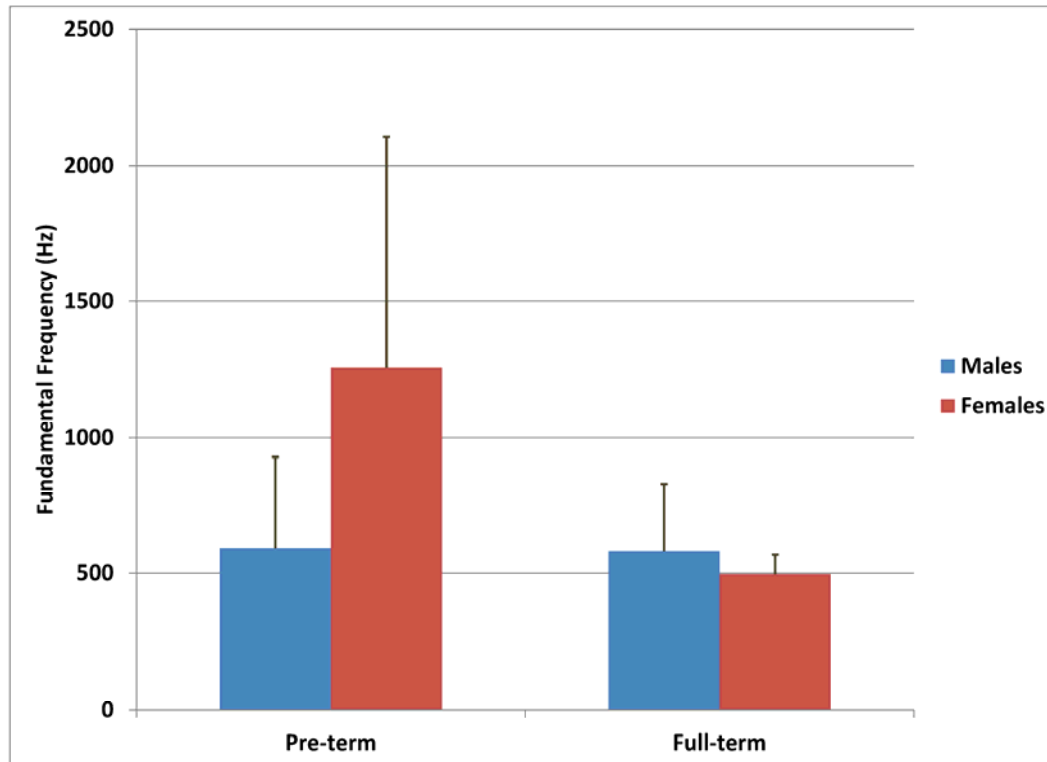
Results

Analysis of First Cry

The mean F0 of the first cry was examined using an Analysis of variance (ANOVA) with between-subjects independent variables of sex (male vs. female), term status (preterm vs. full-term), and position (supine vs. prone). Results indicated a statistically significant main effect

for term status, as preterm infants exhibited a higher mean F0 of the first cry [$F(1,50)=7.37$; $p<0.01$; $\eta^2=.13$]. Additionally, a significant interaction effect for term status by sex was observed [$F(1,50)=10.48$; $p<0.01$; $\eta^2=.17$]. Post-hoc analysis using Bonferroni correction ($\alpha=0.025$) indicated that preterm status was only associated with higher mean F0 of the first cry for female infants [$t(22)=3.48$; $p<0.01$], but not for male infants ($p>0.025$). Figure 1 displays the means of the first cry F0 for pre-term and full-term males and females. All other main effects and interactions were not significant ($p>0.05$).

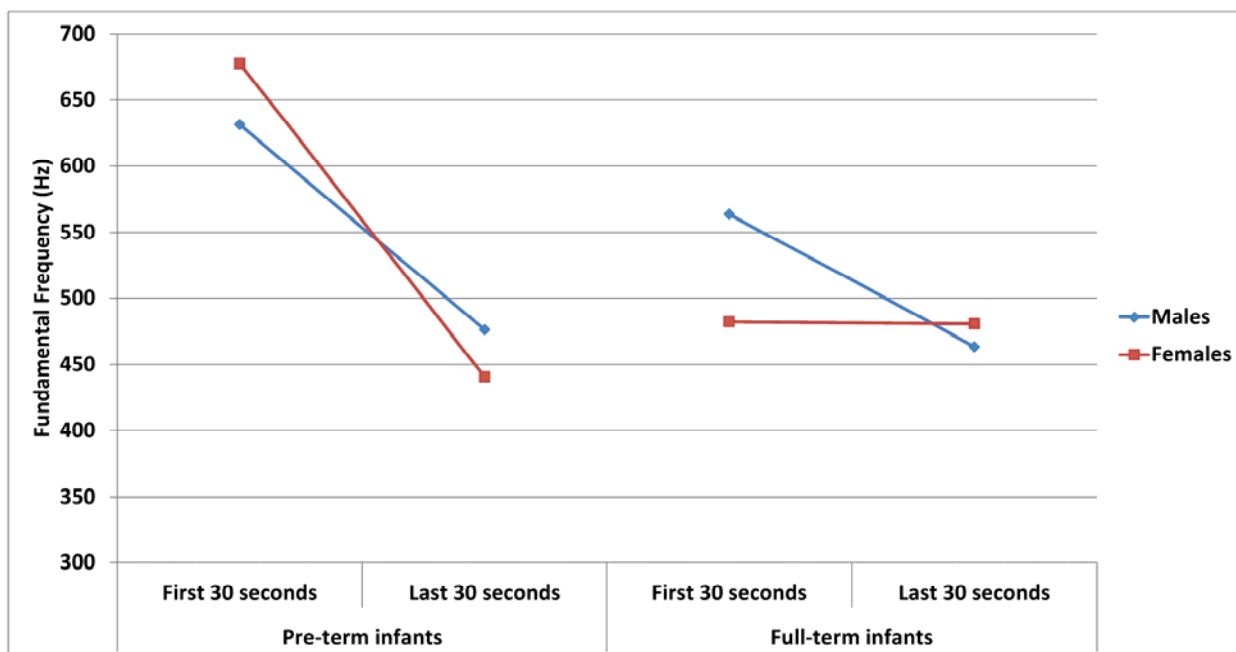
Figure 1. Fundamental Frequency (F0) of the First Cry After the Pain Stimulus is Shown for Both Male and Female Infants and for Both Preterm Infants and Full-Term Infants.



Analysis of Cry Changes Over Time

Researchers compared the mean F0 of the first 30 seconds of the crying episode to the mean F0 of the last 30 seconds of the crying episode (Figure 2). Independent variables were time (early cry vs. late cry), sex (male vs. female), term status (preterm vs. full-term), and position (supine vs. prone). Results from the repeated measures ANOVA indicated a statistically significant main effect for time, with the early portion of the cry having a higher mean F0 than the later portion [$F(1,50)=6.29$; $p=0.02$; $\eta^2=.11$]. Specifically, for the first 30 seconds, Mean F0 was 552.73 Hz (SD=195.31; Range=481.42–677.64 Hz). A significant decrease in Mean F0 is seen in the last 30 seconds compared to the first 30 seconds (last 30 seconds: Mean F0=469.13 Hz; SD=125.67; Range=427.29–629.86 Hz). No statistically significant main effects were found for sex ($p>0.05$), position ($p>0.05$), or term status ($p>0.05$). In addition, there were no significant interaction effects ($p>0.05$).

Figure 2. Fundamental Frequency (F0) of the First 30 Seconds and the Last 30 Seconds of the Cry Episode for Both Male and Female Infants and for Both Preterm and Full-Term Infants.



Discussion

A number of methods have been used to infer infant pain. Johnston and Strada (1986); Craig, Whitfield, Grunau, Linton, and Hadjistavropoulos (1993); and Grunau, Oberlander, Holsti, and Whitfield (1998) all examined infant pain using facial expression, body movement, and heart rate. Other work has included similar measures combined into multidimensional infant pain scales (e.g., Cignacco, Mueller, Hamers, & Gessler, 2004; Hudson-Barr et al., 2002; Krechel & Bildner, 1995; Schade, Joyce, Gerkenmeyer, & Keck, 1996). Additional research has focused on using cry analysis to infer pain. This has included cry studies that were entirely perceptual (e.g., Bellieni et al., 2005; Bellieni et al., 2007), and studies that included acoustic analysis (e.g., Bellieni, Sisto, Cordelli, & Buonocore, 2004; Goberman et al., 2008; Goberman & Robb, 1999; Johnston et al., 1993; Wasz-Hockert, Lind, Vuorenkoski, Partanen, & Valenne, 1968; Zeskind, 1983).

Based on acoustic analysis of adult voices, increased F0 has been found to reflect anxiety / arousal in response to stressful or life-threatening situations (e.g., Goberman, Hughes, & Haydock, 2011; Protopapas & Liberman, 1997; Ruiz, Absil, Harmegnies, Legros, & Poch, 1996). Based on this previous work, this increased anxiety/arousal may cause increased vocal fold tension. The same physiological mechanisms also may relate to infant vocal behavior, as increased F0 has been found to be associated with arousal / pain (Fuller & Horii, 1986).

In the current study, when researchers compared F0 at the beginning 30 seconds to the F0 in the last 30 seconds of the cry episode, they found that the mean F0 decreased over time (regardless of infant sex, term status, or positioning). The current results reflect high levels of tension following the pain stimulus, followed by an overall decrease in tension over time. This finding supports previous findings of a relationship between F0 and pain. During the stress of pain, the laryngeal musculature tightens, effectively raising F0 (Johnston et al., 1993; Wasz-Hockert et al., 1968; Zeskind, 1983). Further, the decrease in F0 from the first 30 seconds to the last 30 seconds may relate to a similar decrease in pain-related tension over time. Previous

researchers examining full crying episodes also found acoustic cry changes over time based on LTAS analysis (Goberman & Robb, 1999; Goberman et al., 2008).

The current study included both preterm and full-term infants. Previous literature has reported higher F0 in preterm compared to full-term infants (Goberman & Robb, 1999; Johnston et al., 1993). Johnston and colleagues (1993) discussed two distinct possibilities for a higher F0 in preterm infant cries. The increase in F0 could relate to either smaller vocal fold mass or to increased vocal tract tension in response to a pain stimulus (Johnson et al., 1993). The current data showed a significantly higher F0 for preterm female infants (compared to preterm males) in the first cry immediately following the pain stimulus. However, the current study did not find any statistically significant F0 difference at the end of the cry episode between preterm infant and full-term infants. Because of this F0 similarity at the end of the cry episode, the data do not support a conclusion related to differences in vocal fold size. Rather, the current data continue to support a relationship between increased tension, F0, and arousal in response to a pain stimulus.

A potentially important application of infant cry analysis is related to the study of SIDS (Goberman et al., 2008). Specifically, one theory states that SIDS may relate to a decreased arousal in response to a life-threatening event in sleep (e.g., American Academy of Pediatrics, Task Force on Sudden Infant Death Syndrome, 2005; Hunt, 2000, 2001). In fact, in the current study decreased arousal also can be inferred from decreased pain cry F0. Specifically, preterm male infants showed decreased F0 arousal and also have previously been shown to be at increased risk for SIDS (Hoffman, Damus, Hillman, & Krongrad, 2006). These data support the continued use of acoustic analysis in the study of SIDS.

Conclusions

The goal of the current study was to examine the F0 of infant cries, with a specific focus on the arousal/laryngeal tension related to a pain response. Overall, the F0 – arousal model may be useful in examining: (a) reaction to pain stimulus, (b) level of arousal in response to pain, (c) decrease in arousal as time since pain increases, and possibly (d) risk for SIDS. The current results further support examination of the first cry following a pain stimulus, along with examination of the entire cry episode.

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